EXPERIMENT - 13

Limitations of Low-Code: Enterprise Case Study

Library Book Issue Management

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Subject	Low-Code Application Development using Zoho Creator
Application	Library Book Issue Management
Platform	Zoho Creator
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Aim

To analyze the limitations of low-code platforms for a complex enterprise library use case and justify a suitable balance between Zoho Creator features and custom services.

Experiment overview

A case-study analysis that extends the student library scenario to a hypothetical multi-campus enterprise and evaluates where additional engineering is required.

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1. Existing System

Library Book Issue Management captures a member request, applies validation, routes it through staff statuses and sends an availability confirmation through a Zoho Mail connection. The current model has request-level category and copy count, rather than a full catalogue of physical book copies.

2. Enterprise Case Assumptions

For analysis only, consider a university network with 10 campuses, 100,000 members and 1,000,000 historical requests. These are hypothetical design targets, not measured capacity or actual project data.

Enterprise need	Why the student model needs extension
Physical-copy inventory	Track title, ISBN, barcode, location and availability for each physical copy.
Concurrent reservations	Two staff members may approve the same last available copy.
Cross-campus transfers	Track source, destination, dispatch and receipt separately.
Reliable notifications	Retry temporary failures without sending duplicate confirmations.
System integration	Connect identity, catalogue and finance systems with stable contracts.
Long-term operations	Monitor latency, retention, recovery, costs and user lifecycle.

3. Method

The study compares these assumed requirements against the implemented library model, the observed trial environment and official platform documentation. It separates application-design gaps from vendor capability limits. No enterprise load test or large dataset was created.

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4. Limitation and Mitigation Analysis

Concern	Risk in this case	Suggested treatment
Data model	Category and quantity cannot identify a physical book.	Add catalogue, physical-copy and loan entities; retain stable IDs.
Concurrency	A read-then-update workflow may approve competing requests.	Use one authoritative reservation operation and test simultaneous requests.
Query workload	Unbounded loops increase application work.	Use criteria, aggregates and bounded pages; measure at scale.
External APIs	Timeouts, quotas and authorization expiry interrupt processing.	Handle responses, use retry limits and record failures without exposing tokens.
Delivery reliability	A sent flag is weaker than end-to-end delivery proof.	Store message identifier/outcome and reconcile bounces or failures.
Customization	Special algorithms or UI behavior may exceed standard components.	Use a reviewed custom function or service boundary.
Governance	Public submission does not prove user identity.	Use authenticated membership for private records and approval actions.
Portability	Schema, data, connections and user mappings are different assets.	Export each needed asset and test import plus reauthorization.
Commercial dependency	Trial and plan entitlements constrain ongoing use.	Review current entitlements before operation; keep independent backups.

These are risks to evaluate, not claims that Creator cannot support enterprise applications. Availability, quotas and performance depend on the chosen plan and architecture.

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5. Proposed Enterprise Architecture

Layer	Responsibility
Zoho Creator	Member/staff forms, reports, approval views and business workflow coordination.
Library domain model	Books, physical copies, members, requests, request lines and loans with stable relationships.
Reservation service	A narrow API that reserves/releases a copy atomically and rejects conflicting requests.
Notification service	Idempotent delivery request, retries, message outcome and audit reference.
Analytics	Aggregated trends with explicit access scope and refresh expectations.
Administration	Identity lifecycle, least-privilege permissions, release history and restore drills.

6. Technology Decision

Keep the current student project in Creator because its form, validation, queue and dashboard requirements fit the platform well. Introduce an external service only when a measured requirement needs it, such as atomic cross-system reservation or specialized computation.

The new Get_Library_Queue_Summary function illustrates a small pro-code extension inside the existing app. Catalyst is a possible future serverless boundary, not a deployed part of this case study.

7. Acceptance Tests for the Proposed Extension

1. Submit two simultaneous reservations for one copy and allow only one success.
2. Retry the same request identifier and confirm that no duplicate loan or email is created.
3. Expire an integration token and verify a recoverable, visible failure.
4. Test permissions with genuinely separate member and staff sessions.
5. Restore a backup into an isolated environment and reconcile record totals.

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8. Result and Conclusion

The case study identifies data modelling, concurrency, integration reliability, governance and portability as the main areas requiring further engineering. The current application is appropriate for the subject project. Enterprise adoption would require additional requirements, load tests and operational controls.

9. Low-Code and Pro-Code Comparison

Approach	Strength	Trade-off
Low-code components	Quick form/report changes and integrated workflows.	Platform behavior and entitlements shape the design.
Custom Deluge function	Reusable domain logic close to app records.	Requires code review, validation and tests.
External custom service	Control over a specialized operation and its interface.	Adds deployment, monitoring and integration maintenance.

10. Evidence Policy

This experiment is an analytical case study. The hypothetical enterprise design is not shown as an implemented system, and no fabricated screenshot is included. Actual implementation evidence remains in Experiments 10, 12 and 14.

References

1. Creator platform performance
2. Creator user permissions
3. Creator functions
4. Catalyst serverless introduction

Official sources checked on 16 October 2026. App observations and test results are identified separately.

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